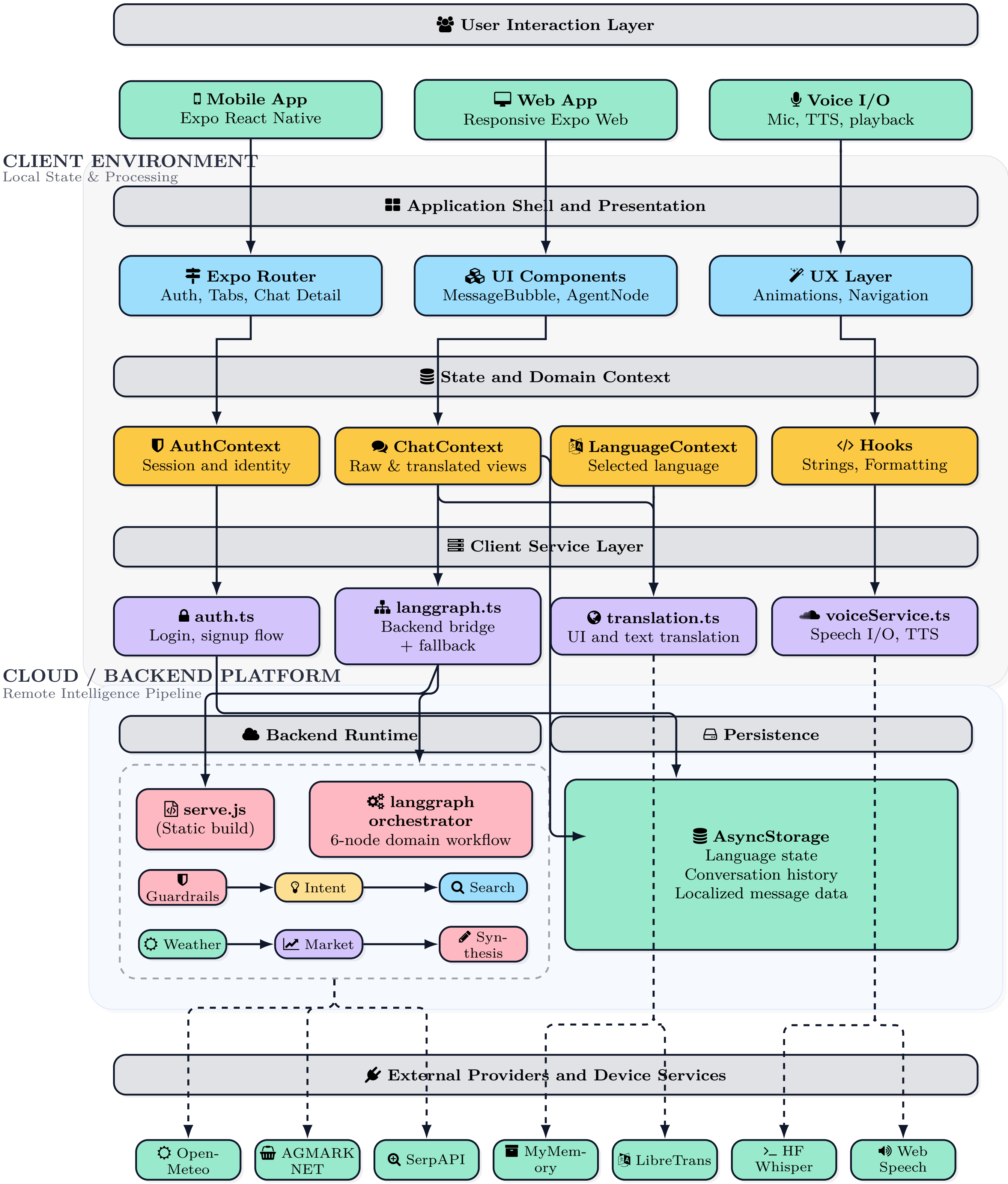
AgriAdvisor Production Architecture

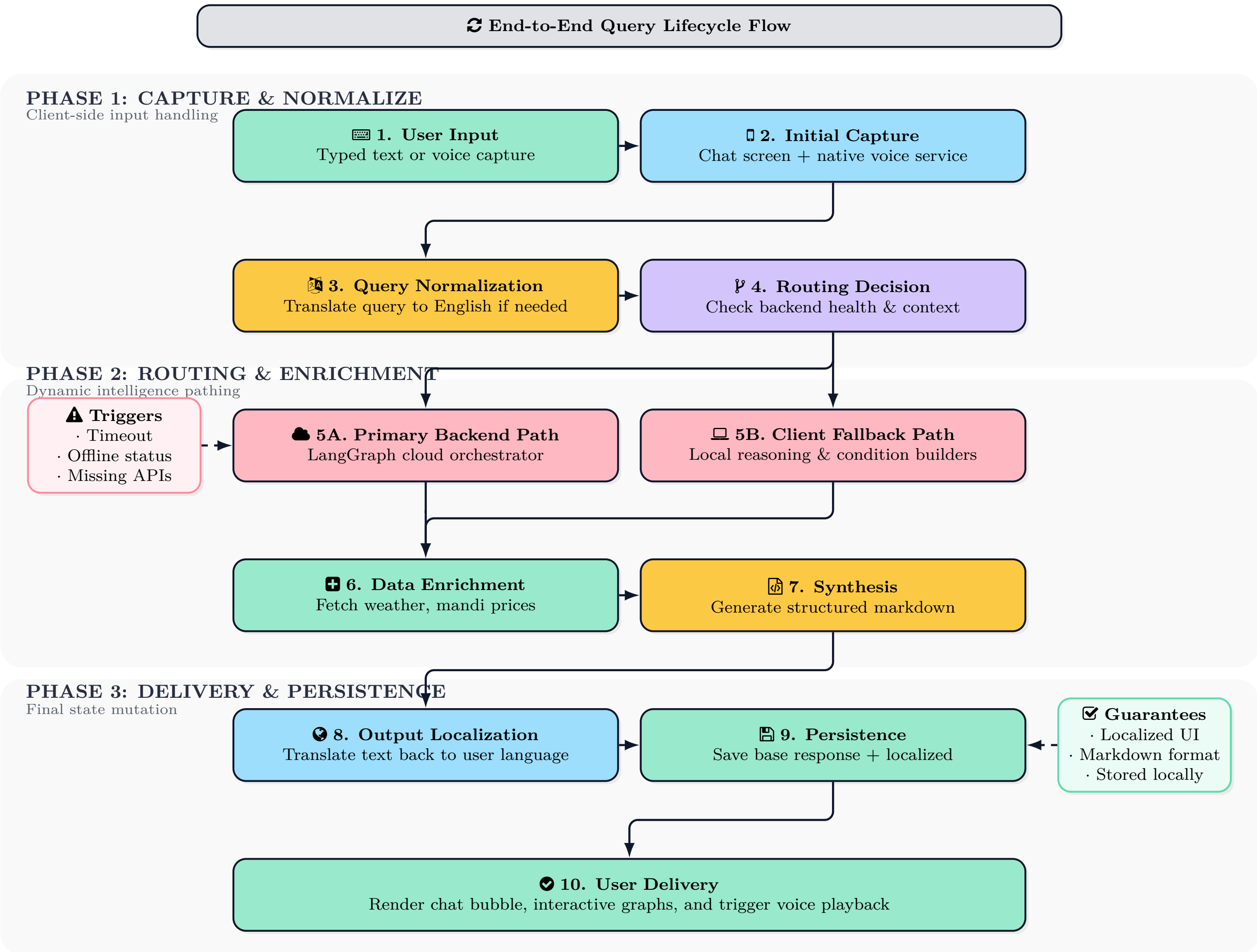
Multilingual mobile and web agricultural advisory platform with resilient orchestration and voice interfaces

Architecture objective. This design documents the current production-oriented architecture, capturing the runtime layers, state ownership, multilingual processing path, fallback behavior, data providers, and voice interfaces for mobile and web.



⇄ Runtime Request and Response Lifecycle

Execution model overview. Every user query initiates a resilient lifecycle designed to guarantee a localized, domain-accurate response. The system first captures input via text or native voice APIs, immediately normalizing the request into an English base for unified processing. Based on backend health and query intent, the orchestrator dynamically routes the request to either the primary LangGraph cloud pipeline or the local client fallback reasoning engine. After enriching the context with live external data, the system synthesizes a structured markdown response. Finally, this payload is translated back to the user's selected language, persisted to local storage alongside the base response, and delivered through the UI and TTS engines.



Component responsibility summary.

Layer	Primary responsibility	Implementation Context
Presentation	Responsive mobile/web UI, chat rendering, graph formatting, voice controls	`app/(tabs)/*`, `MessageBubble.tsx`
State mgt.	Authentication state, language selection, conversation storage	`AuthContext.tsx`, `ChatContext.tsx`
Client logic	Query normalization, backend routing, multilingual translation, speech	`langgraph.ts`, `translation.ts`
Backend API	Domain-safe reasoning workflow with auditable node execution	`langgraph_orchestrator.py`
External APIs	Weather, market data, search, translation, Whisper transcription	Open-Meteo, AGMARKNET, Hugging Face